

SAT Practice Test 1

147 Questions | 3 Modules | Original Content

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MODULE 1 (49 Questions)

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READING & WRITING - MODULE 1

Questions 1-27 | 32 Minutes

Question 1

Skill: Main Idea | Difficulty: Medium

Eighteenth-century Japanese printmaker Kitagawa Utamaro earned wide recognition for his mastery of the *bijin-ga* tradition—portraits of beautiful women. At a time when most *ukiyo-e* artists focused on landscapes or kabuki actors, Utamaro devoted himself to capturing subtle shifts in expression and posture. His commitment to this subject was so influential that later artists who adopted similar techniques were often said to be working “in the manner of Utamaro.”

Which choice best states the main idea of the text?

- A) Utamaro’s focus on landscape prints distinguished him from other *ukiyo-e* artists of his era.
 - B) Utamaro played a significant role in shaping the tradition of portraiture within *ukiyo-e* printmaking.
 - C) Utamaro introduced the *bijin-ga* tradition to Japanese art during the eighteenth century.
 - D) Later artists who imitated Utamaro’s style eventually surpassed his technical achievements.
-

Question 2

Skill: Grammar / Conventions | Difficulty: Medium

The fruit fly *Drosophila melanogaster* has long served as a model organism in genetics research. Because its genome has been fully sequenced and its reproductive cycle is remarkably short, scientists can observe hereditary changes across many generations in a brief period. Both the accessibility of its genetic code and the speed of its life _____ when researchers need to study how specific mutations affect development, they can design experiments that yield results within weeks rather than years.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) cycle matter:
 - B) cycle matter
 - C) cycle, matter
 - D) cycle matter,
-

Question 3

Skill: Quotation Illustrates Claim | Difficulty: Medium

The House of Mirth is a 1905 novel by Edith Wharton. In the novel, the protagonist Lily Bart navigates the rigid social hierarchies of New York's upper class. The narrator portrays Lily as acutely aware of the financial pressures that shape her choices: _____

Which quotation from *The House of Mirth* best illustrates the claim?

- A) "She was so evidently the victim of the civilization which had produced her that the links of her bracelet seemed like manacles chaining her to her fate."
 - B) "The hall was full of the scent of Mrs. Peniston's carnations, which seemed to fill the rooms with their heavy sweetness."
 - C) "Lily smiled at the memories the word evoked, and then turned to look out of the window."
 - D) "The drawing-room curtains were drawn, and the lamps lit, and all the little signs of a well-ordered household were visible."
-

Question 4

Skill: Transitions | Difficulty: Easy

In deep ocean trenches, water pressure can exceed one thousand times atmospheric pressure at sea level. Organisms living at these depths have evolved specialized proteins that maintain their function under extreme compression. _____ the enzymes of deep-sea fish do not simply resist pressure passively; they incorporate structural modifications-such as reduced internal cavities-that actively stabilize their molecular shape.

Which choice completes the text with the most logical transition?

- A) For instance,
 - B) In contrast,
 - C) Regardless,
 - D) As a result,
-

Question 5

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

- The axolotl is an amphibian native to lakes near Mexico City.
- It was believed to be extinct in the wild until a small population was discovered in the canals of Xochimilco in 2013.
- The pangolin is a scaly mammal found in parts of Africa and Asia.
- Pangolins were once thought to be closely related to armadillos but are now classified in their own order, Pholidota.
- Both species face critical threats from habitat loss.

The student wants to specify when the axolotl population was rediscovered. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Native to lakes near Mexico City, axolotls are amphibians that face critical threats from habitat loss.
 - B) A small population of axolotls was discovered in the canals of Xochimilco in 2013, after the species had been thought extinct in the wild.
 - C) Pangolins, once believed to be related to armadillos, are now classified in their own order.
 - D) Both axolotls and pangolins face critical threats from habitat loss and were once thought to be extinct.
-

Question 6

Skill: Main Purpose | Difficulty: Medium

Although Toni Morrison's novels address the lasting psychological effects of slavery and racial injustice, her narratives resist reducing characters to symbols of suffering. She constructs richly layered

inner worlds for her protagonists, granting them desires, contradictions, and moments of joy that exist alongside-and sometimes in tension with-the historical violence they endure. By doing so, Morrison insists that her characters be understood as complete human beings rather than mere illustrations of oppression.

Which choice best states the main purpose of the text?

- A) To outline how Toni Morrison’s novels trace the historical causes of racial injustice in America
 - B) To argue that Toni Morrison’s protagonists are most compelling when they serve as symbols of collective suffering
 - C) To explain how Toni Morrison’s storytelling balances weighty historical themes with full characterization of her protagonists
 - D) To compare Toni Morrison’s approach to characterization with that of other novelists writing about slavery
-

Question 7

Skill: Main Purpose | Difficulty: Easy

In the 1930s, Mexican muralist David Alfaro Siqueiros began experimenting with industrial paints originally designed for automobiles. Unlike traditional oil pigments, these synthetic lacquers dried rapidly and could be applied with spray guns, allowing Siqueiros to cover enormous wall surfaces with vivid, durable color. His adoption of non-traditional materials expanded the possibilities available to public artists and influenced a generation of painters who sought to bring art into everyday urban spaces.

Which choice best states the main purpose of the text?

- A) The text contrasts the artistic philosophies of different Mexican muralists active during the 1930s.
 - B) The text argues that industrial paints are superior to traditional oil pigments in all artistic applications.
 - C) The text describes how Siqueiros’s use of unconventional materials broadened the scope of public mural art.
 - D) The text explains why spray-gun techniques eventually replaced brushwork in professional painting.
-

Question 8

Skill: Overall Structure | Difficulty: Hard

Coral reefs occupy less than one percent of the ocean floor yet support roughly one quarter of all known marine species. Marine biologist Camille Parmesan has noted that even modest increases in sea surface temperature can trigger coral bleaching, a stress response in which corals expel the symbiotic algae that provide them with nutrients and color. Without these algae, corals become vulnerable to disease and may die within weeks. Parmesan argues that protecting reefs therefore requires not only local conservation efforts-such as reducing pollution and overfishing-but also global action to limit greenhouse gas emissions.

Which choice best describes the overall structure of the text?

- A) It identifies an ecological asset, describes a specific threat to that asset, and then outlines the scope of action needed to address the threat.
 - B) It presents an ecological debate, evaluates competing scientific positions, and then sides with the position supported by the most evidence.
 - C) It introduces a controversial conservation policy, critiques the assumptions underlying that policy, and then proposes an alternative.
 - D) It catalogs several unrelated threats facing marine ecosystems and then ranks those threats by severity.
-

Question 9

Skill: Words in Context | Difficulty: Easy

Naguib Mahfouz's *Cairo Trilogy* chronicles three generations of a merchant family against the backdrop of Egypt's struggle for independence. The novels do not simply record historical events; they _____ the emotional and psychological shifts within ordinary households as the outside world transforms.

Which choice completes the text with the most logical and precise word or phrase?

- A) obscure
 - B) trace
 - C) challenge
 - D) predict
-

Question 10

Skill: Words in Context | Difficulty: Easy

The domesticated tomato bears little resemblance to its wild ancestor, a small berry native to western South America. The wild fruit had a tough skin and an intensely tart flavor. Mesoamerican cultivators selectively bred the plant over centuries, gradually _____ the original characteristics until the fruit became the large, thin-skinned, mild-tasting crop familiar today.

Which choice completes the text with the most logical and precise word or phrase?

- A) reinforcing
 - B) cataloging
 - C) altering
 - D) concealing
-

Question 11

Skill: Inference / Complete the Text | Difficulty: Medium

The following text is from Zora Neale Hurston's 1937 novel *Their Eyes Were Watching God*. The protagonist, Janie, reflects on her youth.

She had glossy leaves and bursting buds and she wanted to struggle with life but it seemed to elude her. Where were the singing bees for her? Nothing on the place nor in her grandmama's house answered her. She searched as much of the world as she could from the top of the front steps and then went on down to the front gate and leaned over to gaze up and down the road.

Based on the text, which choice best describes Janie's state of mind?

- A) She feels a restless longing for experiences that her current surroundings cannot provide.
 - B) She is satisfied with the natural beauty that surrounds her grandmother's house.
 - C) She is frightened by the vastness of the world beyond her home.
 - D) She regrets a specific decision she made earlier in her life.
-

Question 12

Skill: Words in Context | Difficulty: Medium

Researchers studying the migration patterns of Arctic terns have found that these birds travel roughly 70,000 kilometers each year between the Arctic and the Antarctic. Elena Gómez and her colleagues tracked 50 individual terns over three consecutive years using lightweight GPS tags. The researchers reported that the terns consistently followed a zigzag route over the Atlantic Ocean rather than a direct north-south path. Gómez hypothesized that this indirect route allows the birds to take advantage of prevailing wind patterns, thereby _____ the energy expenditure of the journey despite increasing its total distance.

Which choice completes the text with the most logical and precise word or phrase?

- A) maximizing
- B) documenting
- C) reducing

D) overlooking

Question 13

Skill: Evidence - Support / Weaken | Difficulty: Hard

Paleontologists have cataloged over thirty distinct tooth shapes among theropod dinosaurs, ranging from blade-like serrated teeth to blunt, peg-shaped structures. Researcher Kenji Yamamoto hypothesized that the diversity of tooth forms corresponds to a similarly broad range of diets, from strict carnivory to herbivory. To test this, Yamamoto analyzed microscopic wear patterns on fossilized teeth and compared them to wear patterns on the teeth of living reptiles with known diets. Although the results confirmed a general link between tooth shape and diet, the analysis also revealed that some theropods with seemingly carnivorous teeth showed wear patterns consistent with a plant-heavy diet.

Which finding, if true, would most directly support the underlined claim?

- A) Several theropod species with serrated teeth exhibited scratch patterns on their enamel that closely matched those found on modern herbivorous iguanas.
 - B) Theropods with peg-shaped teeth were found exclusively in geological formations associated with arid environments.
 - C) Modern crocodilians, which are strict carnivores, display tooth wear patterns that are uniform across all species.
 - D) The number of distinct tooth shapes among theropods increased steadily over the Mesozoic Era.
-

Question 14

Skill: Cross-Text | Difficulty: Medium

Text 1

Literary scholar Amara Singh argues that nineteenth-century travel narratives written by European women should be read primarily as acts of personal liberation. By venturing into unfamiliar territories and recording their experiences, these authors carved out a sphere of intellectual authority that was largely denied to them at home.

Text 2

Historian David Chen cautions that reading women's travel writing solely through the lens of personal empowerment overlooks the ways in which many of these texts reinforced colonial ideologies. Even as female authors challenged gender norms, Chen argues, their descriptions of colonized peoples frequently echoed the same hierarchical assumptions found in writing by their male contemporaries.

Based on the texts, how would Chen (Text 2) most likely respond to Singh's characterization (Text 1) of women's travel narratives?

- A) By arguing that women's travel narratives were less widely read than Singh suggests
 - B) By suggesting that Singh's focus on personal liberation ignores the colonial perspectives embedded in many of these texts
 - C) By agreeing that women's travel narratives were acts of liberation and adding that they also challenged colonial ideologies
 - D) By claiming that male travel writers of the same era were more critical of colonialism than their female counterparts
-

Question 15

Skill: Inference / Complete the Text | Difficulty: Medium

The baobab tree is native to the savannas of sub-Saharan Africa and can live for over a thousand years. Ecologists have noted that baobabs store enormous quantities of water in their spongy trunks—up to 120,000 liters in a single tree. During prolonged dry seasons, elephants strip bark from baobabs to access this moisture. Although the trees can regenerate lost bark over time, repeated stripping by

growing elephant populations has begun to kill some of the oldest specimens. Conservationists therefore argue that protecting baobabs _____

Which choice most logically completes the text?

- A) requires understanding why elephants prefer baobab bark over other water sources in their habitat.
 - B) is unnecessary because the trees' capacity to regenerate bark makes them resilient to any level of damage.
 - C) may depend on managing elephant population density in areas where ancient baobabs are concentrated.
 - D) should focus exclusively on relocating elephants to regions where baobab trees do not grow.
-

Question 16

Skill: Grammar / Conventions | Difficulty: Easy

The phenomenon known as "forest bathing," or *shinrin-yoku*, originated in Japan during the 1980s as a form of nature therapy. Practitioners spend extended periods walking slowly through forested _____ they focus on sensory experiences such as the scent of pine needles and the sound of birdsong.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) areas, where
 - B) areas where,
 - C) areas. Where
 - D) areas where
-

Question 17

Skill: Grammar / Conventions | Difficulty: Easy

Although they operate in vastly different artistic mediums, the sculptor Alberto Giacometti and the playwright Samuel _____ both explored themes of existential isolation and the fragility of human connection in their postwar work.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) Beckett,
 - B) Beckett:
 - C) Beckett;
 - D) Beckett
-

Question 18

Skill: Grammar / Conventions | Difficulty: Medium

In 1928, Scottish bacteriologist Alexander Fleming noticed that a mold colony on a petri dish had killed surrounding bacteria. This chance observation led to the development of _____ what would become one of the most important medical advances of the twentieth century.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) penicillin:
 - B) penicillin,
 - C) penicillin
 - D) penicillin-
-

Question 19

Skill: Grammar / Conventions | Difficulty: Medium

The seeds of the Svalbard Global Seed Vault, a facility built into the permafrost of a Norwegian _____ are kept at a constant temperature of negative eighteen degrees Celsius to preserve genetic material from food crops around the world.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) mountainside,
 - B) mountainside:
 - C) mountainside;
 - D) mountainside
-

Question 20

Skill: Words in Context | Difficulty: Medium

Sandra Cisneros's *The House on Mango Street* is a novella composed of interconnected vignettes. It frequently incorporates Spanish words and phrases into its predominantly English prose. The English sentences surrounding these Spanish elements _____ their meaning through context, so readers unfamiliar with Spanish can follow the narrative without difficulty. Junot Díaz employs a comparable technique in his fiction.

Which choice completes the text with the most logical and precise word or phrase?

- A) contradict
 - B) obscure
 - C) illuminate
 - D) exaggerate
-

Question 21

Skill: Transitions | Difficulty: Medium

Karel Čapek's 1920 play *R.U.R. (Rossum's Universal Robots)* is celebrated for introducing the word "robot" into popular usage, coined from the Czech *robota*, meaning "forced labor." _____ the play also deepened a longstanding tradition in speculative fiction: using invented beings to interrogate what it means to be human.

Which choice completes the text with the most logical transition?

- A) As a consequence of this achievement,
 - B) Regardless of the play's reception,
 - C) In spite of this cultural impact,
 - D) Beyond this linguistic contribution,
-

Question 22

Skill: Inference / Complete the Text | Difficulty: Hard

Researchers studying sleep patterns in dolphins have found that these marine mammals engage in unihemispheric slow-wave sleep, a state in which one half of the brain remains alert while the other half rests. This adaptation allows dolphins to continue swimming and surfacing to breathe even while sleeping. Some scientists have suggested that unihemispheric sleep also enables dolphins to maintain vigilance against predators, but marine biologist Lena Johansson points out that dolphins in captivity—where predator threats are absent—exhibit the same sleep pattern, suggesting that _____

Which choice most logically completes the text?

- A) the need to breathe at the surface, rather than predator avoidance, is the primary reason dolphins evolved this form of sleep.
- B) dolphins in captivity sleep significantly more hours per day than dolphins in the wild.
- C) unihemispheric sleep is unique to dolphins and has not been observed in other marine mammals.
- D) predator avoidance is a more important survival function than breathing for most marine mammals.

Question 23

Skill: Inference / Complete the Text | Difficulty: Medium

The following text is adapted from Nella Larsen's 1929 novel *Passing*. The narrator, Irene, encounters an acquaintance she has not seen in years.

She stared at the woman before her, seized by an odd sense of both familiarity and estrangement. The face was unmistakably the same-the same high cheekbones, the same watchful eyes-and yet something fundamental had shifted. It was not merely the passage of time; it was as though the woman had rebuilt herself from the inside, adopting a new posture, a new cadence of speech, a new way of occupying space.

Based on the text, what does Irene observe about the woman?

- A) The woman's physical features have changed dramatically since their last meeting.
- B) The woman appears to be deliberately concealing her identity from Irene.
- C) Irene cannot remember where she previously met the woman.
- D) While the woman looks the same, her manner and self-presentation have fundamentally changed.

Question 24

Skill: Grammar / Conventions | Difficulty: Hard

Although quinoa is now cultivated on several continents, the crop was first domesticated in the Andean highlands roughly 5,000 years ago. A 2023 genomic study led by botanist Priya Nair found that modern quinoa cultivars retain approximately 85 percent of the genetic diversity present in their wild _____ a finding that surprised researchers, since most domesticated grains have lost far more diversity through selective breeding.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) ancestors,
- B) ancestors:
- C) ancestors-
- D) ancestors

Question 25

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

- The Trans-Siberian Railway spans approximately 9,289 kilometers from Moscow to Vladivostok.
- Construction began in 1891 and was completed in 1916.
- It remains the longest railway line in the world.
- The Qinghai-Tibet Railway, completed in 2006, is the highest railway in the world.
- It reaches a maximum elevation of 5,072 meters above sea level.

The student wants to emphasize the Trans-Siberian Railway's record-setting length and provide historical context. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) At 5,072 meters above sea level, the Qinghai-Tibet Railway is the highest railway in the world.
- B) Spanning approximately 9,289 kilometers from Moscow to Vladivostok, the Trans-Siberian Railway-completed in 1916 after 25 years of construction-remains the longest railway line in the world.
- C) Both the Trans-Siberian Railway and the Qinghai-Tibet Railway hold world records related to their size.
- D) The Trans-Siberian Railway, which is approximately 9,289 kilometers long, is considerably longer than the Qinghai-Tibet Railway.

Question 26

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

- Ada Lovelace wrote what many historians consider the first computer program in 1843.
- The program was designed for Charles Babbage’s proposed Analytical Engine.
- The Analytical Engine was never built during Babbage’s lifetime.
- Lovelace’s notes include a detailed description of how the engine could compute Bernoulli numbers.
- Historians debate whether Lovelace or Babbage deserves primary credit for the algorithm.

The student wants to acknowledge a complication in attributing the first computer program. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Ada Lovelace wrote a program in 1843 for a machine designed by Charles Babbage called the Analytical Engine.
- B) The Analytical Engine, which Charles Babbage proposed but never built, was designed to perform complex calculations.
- C) Although Ada Lovelace’s 1843 notes describe a detailed algorithm for computing Bernoulli numbers on the Analytical Engine, historians continue to debate whether Lovelace or Babbage deserves primary credit for the work.
- D) Lovelace’s program was designed to compute Bernoulli numbers, a sequence of rational numbers important in number theory.

Question 27

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

- Wangari Maathai (1940–2011) was a Kenyan environmental activist and Nobel Peace Prize laureate.
- She founded the Green Belt Movement in 1977, which organized rural women to plant trees to combat deforestation.
- Under her leadership, the movement planted over 51 million trees across Kenya.
- The Green Belt Movement also advocated for women’s rights and democratic governance.
- Contemporary geographer Imani Osei has described Maathai’s work as “an example of how environmental activism and social justice can be pursued as a single, unified cause.”

The student wants to describe Maathai’s approach and explain its broader significance. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Wangari Maathai founded the Green Belt Movement in 1977, and the organization has planted over 51 million trees in Kenya.
- B) According to geographer Imani Osei, environmental activism and social justice are related causes.
- C) Through the Green Belt Movement, Wangari Maathai organized rural Kenyan women to plant trees while also advocating for women’s rights—an approach that geographer Imani Osei has cited as evidence that environmental and social justice work can be pursued as a unified cause.
- D) Wangari Maathai won the Nobel Peace Prize for her work with the Green Belt Movement, an organization that planted over 51 million trees in Kenya.

MATH - MODULE 1

Questions 28–49 | 35 Minutes

Directions: For multiple-choice questions, choose the best answer from the options provided. For student-produced response questions, enter your answer in the grid provided. You may use a calculator for all questions in this section.

Question 28

Skill: Systems of Linear Equations | Difficulty: Medium

A bookstore sells novels for \$14 each and poetry collections for \$9 each. If Marta spent exactly \$100 buying a combination of novels and poetry collections, and she bought at least one of each, how many novels did she buy?

- A) 1
 - B) 2
 - C) 4
 - D) 5
-

Question 29

Skill: Data Interpretation from Tables and Graphs | Difficulty: Easy

GRAPH TO INSERT

A research team recorded the number of bird species observed at four different wetland sites.

Site	Area (hectares)	Species observed
Marshfield	12	34
Cedar Lake	18	45
Pine Bog	8	24
Heron Flats	25	58

Based on the table, which site had the greatest number of species observed per hectare?

- A) Marshfield
 - B) Cedar Lake
 - C) Pine Bog
 - D) Heron Flats
-

Question 30

Skill: Function Notation & Evaluation | Difficulty: Easy

The function f is defined by $f(x) = 3x + 7$. What is the value of $f(5)$?

- A) 15
 - B) 57
 - C) 35
 - D) 22
-

Question 31

Skill: Graphs of Linear Equations | Difficulty: Easy

A water tank is being filled at a constant rate. At time $t = 0$ minutes, the tank contains 15 gallons. At time $t = 6$ minutes, the tank contains 39 gallons. At what rate, in gallons per minute, is the tank being filled?

- A) 3
- B) 4
- C) 5

D) 6

Question 32

Skill: Linear Equations in One & Two Variables | Difficulty: Easy

If $3(2x - 4) = 18$, what is the value of x ?

- A) 3
 - B) 9
 - C) 7
 - D) 5
-

Question 33

Skill: Percentages | Difficulty: Easy

A survey of 200 students found that 45% preferred reading fiction. How many students preferred reading fiction?

- A) 45
 - B) 80
 - C) 90
 - D) 110
-

Question 34

Skill: Linear & Exponential Models from Data | Difficulty: Easy

GRAPH TO INSERT

The scatterplot below shows the relationship between the number of hours studied and the test score for 15 students. A line of best fit for the data has a slope of 8 and a y-intercept of 40.

Based on the line of best fit, what is the predicted test score for a student who studies for 6 hours?

- A) 48
 - B) 78
 - C) 82
 - D) 88
-

Question 35

Skill: Exponent Rules & Properties | Difficulty: Easy

Which expression is equivalent to $(4x^3)(5x^2)$?

- A) $9x^5$
 - B) $9x^6$
 - C) $20x^5$
 - D) $20x^6$
-

Question 36

Skill: Area & Perimeter | Difficulty: Easy

A rectangle has a length that is 3 times its width. If the perimeter of the rectangle is 64 centimeters, what is the width, in centimeters?

- A) 8
- B) 16
- C) 24
- D) 32

Question 37

Skill: Ratios, Rates & Proportions | Difficulty: Easy

In a bag containing only red and blue marbles, the ratio of red marbles to blue marbles is 3:5. If there are 40 marbles in the bag, how many are red?

- A) 8
 - B) 12
 - C) 15
 - D) 25
-

Question 38

Skill: Graphs of Linear Equations | Difficulty: Easy

Line m passes through the points (2, 5) and (6, 13). What is the slope of line m ?

- A) $\frac{1}{2}$
 - B) 8
 - C) 4
 - D) 2
-

Question 39

Skill: Percentages | Difficulty: Medium

A store reduces the price of a jacket by 20%. If the sale price is \$56, what was the original price of the jacket?

- A) \$44.80
 - B) \$67.20
 - C) \$70.00
 - D) \$76.00
-

Question 40

Skill: Systems of Linear Equations | Difficulty: Easy

What is the solution to the system of equations?

$$2x + y = 11$$

$$x - y = 1$$

- A) $x = 3, y = 5$
 - B) $x = 6, y = -1$
 - C) $x = 5, y = 1$
 - D) $x = 4, y = 3$
-

Question 41

Skill: Area & Perimeter | Difficulty: Easy

A triangle has a base of 10 inches and a height of 7 inches. What is the area of the triangle, in square inches?

- A) 17
- B) 35
- C) 70
- D) 140

Question 42

Skill: Creating Linear Models from Word Problems | Difficulty: Easy

If a car travels at a constant speed of 55 miles per hour, which equation represents the distance d , in miles, the car travels in t hours?

- A) $d = t + 55$
 - B) $d = 55/t$
 - C) $d = t/55$
 - D) $d = 55t$
-

Question 43

Skill: Statistics: Mean & Median | Difficulty: Easy

The mean of five numbers is 12. Four of the numbers are 8, 10, 14, and 16. What is the fifth number?

- A) 10
 - B) 12
 - C) 14
 - D) 15
-

Question 44

Skill: Volume & Surface Area | Difficulty: Medium

A cylindrical water bottle has a radius of 4 centimeters and a height of 20 centimeters. What is the volume of the water bottle, in cubic centimeters? (Volume of a cylinder = $\pi r^2 h$)

- A) 80π
 - B) 160π
 - C) 320π
 - D) $1,280\pi$
-

Question 45

Skill: Probability - Basic | Difficulty: Medium

In a class of 30 students, 18 study French and 12 study Spanish. If 5 students study both languages, how many students study at least one of the two languages?

- A) 20
 - B) 25
 - C) 27
 - D) 30
-

Question 46

Skill: Data Interpretation from Tables and Graphs | Difficulty: Medium

GRAPH TO INSERT

A marine biologist recorded the average weight (in grams) of four species of tropical fish in two different habitats.

Species	Reef habitat (g)	Open water habitat (g)
Parrotfish	320	280
Wrasse	150	145
Damselfish	45	40
Surgeonfish	200	190

The biologist notes that the two species with the most similar average weights across the two habitats are _____

Which choice most effectively uses data from the table to complete the statement?

- A) Parrotfish and Surgeonfish.
- B) Wrasse and Surgeonfish.
- C) Damselfish and Surgeonfish.
- D) Wrasse and Damselfish.

Question 47

Skill: Unit Conversion & Dimensional Analysis | Difficulty: Medium

A farmer plants corn in rows. Each row is 120 feet long, and the farmer plants one seed every 8 inches. How many seeds does the farmer plant in one row? (1 foot = 12 inches)

- A) 15
- B) 150
- C) 180
- D) 1,440

Question 48

Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Easy

Student-Produced Response (Grid-In)

If $4x + 7 = 31$, what is the value of $8x + 14$?

Question 49

Skill: Systems of Linear Equations | Difficulty: Medium

Student-Produced Response (Grid-In)

A bakery sells cupcakes in boxes of 6 and boxes of 10. If a customer buys exactly 3 boxes and receives exactly 22 cupcakes, how many boxes of 6 did the customer buy?

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MODULE 2 - EASY ADAPTIVE (49 Questions)

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READING & WRITING - EASY MODULE

Questions 50-76 | 32 Minutes

Question 50

Skill: Main Idea | Difficulty: Easy

The migration of monarch butterflies is one of the most remarkable journeys in the animal kingdom. Each autumn, millions of monarchs travel up to 4,800 kilometers from Canada and the United States to forests in central Mexico. Scientists have found that the butterflies use a combination of the sun's position and Earth's magnetic field to navigate across such vast distances.

Which choice best states the main idea of the text?

- A) Monarch butterflies migrate long distances using sophisticated navigational methods.
 - B) Monarch butterflies are the only insect species capable of long-distance migration.
 - C) Scientists disagree about how monarch butterflies navigate during migration.
 - D) The forests of central Mexico are the only suitable habitat for monarch butterflies.
-

Question 51

Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:

- The Nobel Prize in Literature is awarded annually by the Swedish Academy.
- The academy recognizes writers who have produced outstanding work in an idealistic direction.
- Wole Soyinka became the first African writer to win the prize in 1986.
- Soyinka is a Nigerian playwright, poet, and essayist.

The student wants to identify who Wole Soyinka is and explain the significance of his award. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The Nobel Prize in Literature is awarded annually by the Swedish Academy to writers who have produced outstanding work.
 - B) In 1986, Nigerian playwright, poet, and essayist Wole Soyinka became the first African writer to win the Nobel Prize in Literature.
 - C) Wole Soyinka is a Nigerian writer who has produced plays, poems, and essays.
 - D) The Swedish Academy awarded the Nobel Prize in Literature to an African writer in 1986.
-

Question 52

Skill: Words in Context | Difficulty: Easy

The painter Georgia O'Keeffe is best known for her large-scale flower paintings, which _____ tiny botanical details into monumental compositions that fill entire canvases.

Which choice completes the text with the most logical and precise word or phrase?

- A) diminish
 - B) overlook
 - C) misrepresent
 - D) transform
-

Question 53

Skill: Transitions | Difficulty: Easy

In her 2019 study, archaeologist Lucia Vega examined pottery fragments found at a site in northern Peru. The fragments displayed decorative patterns that had not been previously documented in the region. _____ Vega proposed that the pottery was produced by a local workshop rather than imported from a distant culture.

Which choice completes the text with the most logical transition?

- A) Nevertheless,
- B) In other words,
- C) On the basis of this evidence,
- D) In spite of these findings,

Question 54

Skill: Inference / Complete the Text | Difficulty: Medium

The following text is from Louisa May Alcott's 1868 novel *Little Women*. The character Jo is described during a moment at home.

Jo immediately sat up, put her hands in her pockets, and began to whistle. "Don't, Jo; it's so boyish." "That's why I do it."

Based on the text, what can most reasonably be inferred about Jo?

- A) She deliberately adopts behaviors that challenge conventional expectations.
 - B) She is unaware that her behavior is considered unusual.
 - C) She regrets her habit of whistling but cannot stop.
 - D) She whistles primarily to entertain her family members.
-

Question 55

Skill: Transitions | Difficulty: Easy

A museum exhibit describes the history of the printing press. The exhibit notes that Johannes Gutenberg developed movable type in Europe around 1440. _____ his invention built upon earlier printing technologies that had been used in East Asia for centuries.

Which choice completes the text with the most logical transition?

- A) Therefore,
 - B) Similarly,
 - C) However,
 - D) In conclusion,
-

Question 56

Skill: Grammar / Conventions | Difficulty: Easy

Polar bears are classified as marine mammals because they depend on sea ice as a platform for hunting seals. As Arctic temperatures rise and sea ice _____ polar bears must swim longer distances between ice floes, which increases their energy expenditure.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) recedes;
 - B) recedes:
 - C) recedes
 - D) recedes,
-

Question 57

Skill: Grammar / Conventions | Difficulty: Easy

The city of Venice, Italy, is built on a network of 118 small islands connected by approximately 400 _____. _____ the city's unusual geography has made it a popular destination for tourists from around the world.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) bridges, and
 - B) bridges and,
 - C) bridges. And
 - D) bridges and
-

Question 58

Skill: Grammar / Conventions | Difficulty: Easy

Musician and composer Duke Ellington led his jazz orchestra for nearly fifty years. His compositions, which blended elements of blues, gospel, and classical _____ are considered among the most significant contributions to American music.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) music,
 - B) music
 - C) music;
 - D) music-
-

Question 59

Skill: Grammar / Conventions | Difficulty: Easy

The *Odyssey*, an epic poem attributed to the ancient Greek poet Homer, tells the story of Odysseus's long journey home after the Trojan War. Throughout the _____ Odysseus encounters supernatural creatures and divine beings that both help and hinder his progress.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) narrative. Odysseus
 - B) narrative; Odysseus
 - C) narrative,
 - D) narrative
-

Question 60

Skill: Grammar / Conventions | Difficulty: Medium

A bias toward studying ecosystems at a regional rather than a global scale was one of the many limitations ecologist Marco Torres and his team _____ reviewing over 300 biodiversity studies from six continents.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) discovered, while
 - B) discovered while
 - C) discovered: while
 - D) discovered. While
-

Question 61

Skill: Grammar / Conventions | Difficulty: Easy

With an average annual rainfall of only 254 millimeters, the Atacama Desert in Chile is one of the driest _____ on Earth that scientists have studied for clues about the possibility of life on Mars.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) places
 - B) places,
 - C) places;
 - D) places-
-

Question 62

Skill: Words in Context | Difficulty: Easy

The practice of crop rotation-alternating which crops are planted in a field from season to season-helps maintain soil fertility. Farmers who grow the same crop year after year often find that the soil becomes depleted of specific nutrients. By rotating between nitrogen-fixing legumes and nutrient-demanding grains, farmers can _____ the need for synthetic fertilizers.

Which choice completes the text with the most logical and precise word or phrase?

- A) increase
 - B) ignore
 - C) reduce
 - D) misunderstand
-

Question 63

Skill: Transitions | Difficulty: Easy

Geologists studying volcanic eruptions on the island of Iceland have developed early warning systems that detect subtle changes in ground elevation. These shifts, sometimes measuring only a few centimeters, can indicate that magma is rising beneath the surface. _____ the detection systems have provided residents with critical time to evacuate before several recent eruptions.

Which choice completes the text with the most logical transition?

- A) Conversely,
 - B) As a result,
 - C) Regardless,
 - D) In contrast,
-

Question 64

Skill: Inference / Complete the Text | Difficulty: Medium

The following text is from Charlotte Brontë's 1847 novel *Jane Eyre*. The narrator describes the landscape during a walk.

The ground was hard, the air was still, my road was lonely; I walked fast till I got warm, and then I walked slowly to enjoy and analyse the species of pleasure brooding for me in the hour and situation.

Based on the text, what can most reasonably be inferred about the narrator?

- A) She finds satisfaction in solitary moments spent observing her surroundings.
 - B) She is walking quickly because she is late for an important appointment.
 - C) She is uncomfortable being alone and wishes for company.
 - D) She is lost and trying to find the correct road.
-

Question 65

Skill: Words in Context | Difficulty: Easy

The following text is from Chimamanda Ngozi Adichie's 2013 novel *Americanah*. The narrator describes arriving at a new apartment.

She set down her suitcase and looked around the small room. The walls were a cheerful yellow, and a narrow window let in a strip of afternoon light. It was not much, but it was hers, entirely hers, and that knowledge settled over her like a warm cloth.

As used in the text, what does the phrase "settled over her" most nearly mean?

- A) Weighed her down
- B) Enveloped her gently
- C) Startled her
- D) Confused her

Question 66

Skill: Main Idea | Difficulty: Easy

Researchers studying honeybee communication have found that worker bees perform a “waggle dance” to indicate the direction and distance of food sources to other members of the hive. The angle of the dance relative to the vertical surface of the honeycomb corresponds to the angle between the food source and the sun.

Which choice best states the main idea of the text?

- A) Honeybees use a specific physical movement to share navigational information with their hive.
- B) The waggle dance is performed only by queen bees to direct worker bees.
- C) Honeybees are the only insects that communicate through physical movement.
- D) The direction of the sun has no effect on honeybee navigation.

Question 67

Skill: Words in Context | Difficulty: Easy

The city of Cusco, Peru, served as the capital of the Inca Empire from the thirteenth to the sixteenth century. Many of the massive stone walls built by Inca engineers still stand today, a testament to the precision of their construction. The stones were cut and fitted together so tightly that not even a knife blade can be _____ between them.

Which choice completes the text with the most logical and precise word or phrase?

- A) concealed
- B) inserted
- C) launched
- D) imagined

Question 68

Skill: Words in Context | Difficulty: Easy

Marine biologists have observed that clownfish and sea anemones engage in a mutually beneficial relationship. The anemone’s stinging tentacles protect the clownfish from predators, while the clownfish attracts prey that the anemone then _____.

Which choice completes the text with the most logical and precise word or phrase?

- A) avoids
- B) investigates
- C) releases
- D) consumes

Question 69

Skill: Grammar / Conventions | Difficulty: Medium

Renowned architect Zaha Hadid designed buildings characterized by sweeping curves and angular _____ her distinctive style challenged conventional notions of what a building could look like.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) geometries,
 - B) geometries and
 - C) geometries
 - D) geometries;
-

Question 70

Skill: Cross-Text | Difficulty: Medium

Text 1

Traditional farming methods, such as terracing and intercropping, have sustained agricultural communities for thousands of years. Agronomist Elena Ruiz argues that these practices are inherently sustainable because they work with natural ecosystems rather than against them.

Text 2

Agricultural scientist James Park contends that while traditional methods offer valuable ecological insights, they cannot alone meet the food demands of a growing global population. Park suggests that combining traditional practices with modern technology-such as precision irrigation and soil sensors-offers the most practical path forward.

Based on the texts, how would Park (Text 2) most likely respond to Ruiz's argument (Text 1)?

- A) By rejecting the claim that traditional farming methods are sustainable in any context
- B) By agreeing that traditional methods are valuable but arguing they must be supplemented with modern technology to meet current demands
- C) By suggesting that precision irrigation is more sustainable than terracing
- D) By claiming that traditional farming methods have already been abandoned worldwide

Question 71

Skill: Main Purpose | Difficulty: Easy

The ancient city of Petra, carved into sandstone cliffs in present-day Jordan, served as the capital of the Nabataean kingdom. The city's most famous structure, Al-Khazneh (the Treasury), features a façade over 40 meters tall adorned with columns and sculptures. Archaeologists believe that the structure served as a royal tomb rather than an actual treasury.

Which choice best states the main purpose of the text?

- A) To describe a notable structure in Petra and clarify its likely original function
- B) To compare the architecture of Petra with that of other ancient cities
- C) To argue that archaeologists have misidentified the purpose of Al-Khazneh
- D) To explain how the Nabataean kingdom acquired its wealth

Question 72

Skill: Transitions | Difficulty: Easy

When the body is exposed to cold temperatures, blood vessels near the skin constrict to reduce heat loss. This response, known as vasoconstriction, redirects blood flow toward the body's core organs. _____ the extremities-hands, feet, and ears-receive less blood and become more vulnerable to frostbite.

Which choice completes the text with the most logical transition?

- A) Similarly,
- B) Consequently,
- C) Alternatively,
- D) Previously,

Question 73

Skill: Words in Context | Difficulty: Easy

Inventor Nikola Tesla developed the alternating current (AC) electrical system that powers most homes and businesses today. His work on AC technology was initially _____ by Thomas Edison, who promoted the competing direct current (DC) system, but Tesla's design ultimately proved more efficient for transmitting electricity over long distances.

Which choice completes the text with the most logical and precise word or phrase?

- A) endorsed
 - B) ignored
 - C) replicated
 - D) opposed
-

Question 74

Skill: Grammar / Conventions | Difficulty: Easy

The world's largest flower, *Rafflesia arnoldii*, can measure up to one meter in diameter. Found in the rainforests of Southeast _____ the plant has no leaves, stems, or roots of its own and survives entirely as a parasite on tropical vines.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) Asia;
 - B) Asia
 - C) Asia,
 - D) Asia-
-

Question 75

Skill: Quotation Illustrates Claim | Difficulty: Medium

The Wind in the Willows is a 1908 novel by Kenneth Grahame. In the story, Mole abandons his underground home on a spring morning and ventures to the riverbank, where he meets the Water Rat. The narrator describes Mole as captivated by the unfamiliar world above ground: _____

Which quotation from *The Wind in the Willows* best illustrates the claim?

- A) "The Mole had been working very hard all the morning, spring-cleaning his little home."
 - B) "He thought his happiness was complete when, as he meandered aimlessly along, suddenly he stood by the edge of a full-fed river."
 - C) "The Rat said nothing, but stooped and unfastened a rope and hauled on it."
 - D) "'Nice? It's the only thing,' said the Water Rat solemnly."
-

Question 76

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

- The Great Barrier Reef is the largest coral reef system in the world.
- It stretches over 2,300 kilometers along the northeast coast of Australia.
- The reef is home to more than 1,500 species of fish.
- It was designated a UNESCO World Heritage Site in 1981.
- Coral bleaching events in 2016 and 2017 damaged approximately 50% of the reef.

The student wants to present the reef's ecological importance alongside a current threat. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The Great Barrier Reef, a UNESCO World Heritage Site since 1981, stretches over 2,300 kilometers along the coast of Australia.
 - B) Although the Great Barrier Reef supports over 1,500 species of fish, coral bleaching events in 2016 and 2017 damaged roughly half of the reef system.
 - C) The Great Barrier Reef is the largest coral reef system in the world and is located off the northeast coast of Australia.
 - D) Coral bleaching events have affected many reefs around the world, including the Great Barrier Reef.
-

MATH - EASY MODULE

Questions 77-98 | 35 Minutes

Question 77

Skill: Function Notation & Evaluation | Difficulty: Easy

What is the value of $3x - 5$ when $x = 4$?

- A) 2
 - B) 17
 - C) 12
 - D) 7
-

Question 78

Skill: Percentages | Difficulty: Easy

A school has 250 students. If 60% of the students are enrolled in at least one extracurricular activity, how many students are enrolled in at least one extracurricular activity?

- A) 60
 - B) 100
 - C) 150
 - D) 190
-

Question 79

Skill: Statistics: Mean & Median | Difficulty: Easy

The table below shows the number of books read by four students during the summer.

Student	Books read
Aisha	7
Ben	12
Carla	5
David	8

What is the mean number of books read by the four students?

- A) 7
 - B) 7.5
 - C) 8.5
 - D) 8
-

Question 80

Skill: Linear Equations in One & Two Variables | Difficulty: Easy

If $2x + 3 = 15$, what is the value of x ?

- A) 3
 - B) 6
 - C) 9
 - D) 12
-

Question 81

Skill: Ratios, Rates & Proportions | Difficulty: Easy

A recipe calls for 3 cups of flour to make 24 cookies. How many cups of flour are needed to make 40 cookies?

- A) 3.5
 - B) 4
 - C) 6
 - D) 5
-

Question 82

Skill: Area & Perimeter | Difficulty: Easy

What is the area of a rectangle with a length of 12 meters and a width of 5 meters?

- A) 17 square meters
 - B) 34 square meters
 - C) 60 square meters
 - D) 120 square meters
-

Question 83

Skill: Probability - Basic | Difficulty: Easy

A bag contains 3 red balls and 7 blue balls. If one ball is selected at random, what is the probability that it is red?

- A) $\frac{3}{10}$
 - B) $\frac{7}{10}$
 - C) $\frac{3}{7}$
 - D) $\frac{7}{3}$
-

Question 84

Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Easy

Which of the following is equivalent to $3(x + 4) - 2x$?

- A) $x + 4$
 - B) $x + 12$
 - C) $5x + 4$
 - D) $5x + 12$
-

Question 85

Skill: Ratios, Rates & Proportions | Difficulty: Easy

A train travels at a constant speed of 80 kilometers per hour. How far does the train travel in 2.5 hours?

- A) 160 kilometers
 - B) 180 kilometers
 - C) 200 kilometers
 - D) 240 kilometers
-

Question 86

Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Easy

On a map, 1 centimeter represents 50 kilometers. If two cities are 7 centimeters apart on the map, what is the actual distance between the cities?

- A) 57 kilometers
 - B) 200 kilometers
 - C) 300 kilometers
 - D) 350 kilometers
-

Question 87

Skill: Graphs of Linear Equations | Difficulty: Easy

GRAPH TO INSERT

The graph of a linear function passes through the origin and the point (4, 12). What is the slope of the line?

- A) $\frac{1}{3}$
 - B) 3
 - C) 4
 - D) 12
-

Question 88

Skill: Percentages | Difficulty: Easy

If a shirt originally costs \$40 and is on sale for 25% off, what is the sale price?

- A) \$30
 - B) \$15
 - C) \$10
 - D) \$35
-

Question 89

Skill: Area & Perimeter | Difficulty: Medium

A square garden has a perimeter of 36 feet. What is the area of the garden, in square feet?

- A) 9
 - B) 36
 - C) 81
 - D) 144
-

Question 90

Skill: Linear Inequalities | Difficulty: Easy

Which value of x satisfies the inequality $2x - 3 > 7$?

- A) 3
 - B) 4
 - C) 5
 - D) 6
-

Question 91

Skill: Statistics: Mean & Median | Difficulty: Easy

The median of the data set {2, 5, 7, 8, 13, 15, 20} is:

- A) 7
- B) 8

- C) 10
 - D) 13
-

Question 92

Skill: Creating Linear Models from Word Problems | Difficulty: Medium

A parking lot charges \$3 for the first hour and \$2 for each additional hour. Which expression represents the total cost, in dollars, for parking h hours, where $h > 1$?

- A) $3 + 2(h - 1)$
 - B) $5h$
 - C) $3 + 2h$
 - D) $2 + 3(h - 1)$
-

Question 93

Skill: Coordinate Geometry | Difficulty: Easy

Point A is at coordinates (1, 3) and point B is at coordinates (5, 3). What is the distance between points A and B?

- A) 2
 - B) 3
 - C) 6
 - D) 4
-

Question 94

Skill: Linear Equations in One & Two Variables | Difficulty: Easy

If $5y = 35$, what is the value of $y + 2$?

- A) 5
 - B) 7
 - C) 9
 - D) 12
-

Question 95

Skill: Circle Equations | Difficulty: Easy

A circular pool has a diameter of 10 meters. What is the circumference of the pool, in meters? (Use $C = \pi d$)

- A) 5π
 - B) 10π
 - C) 25π
 - D) 100π
-

Question 96

Skill: Percentages | Difficulty: Medium

Student-Produced Response (Grid-In)

A number is increased by 40% and the result is 56. What is the original number?

Question 97

Skill: Linear Equations in One & Two Variables | Difficulty: Easy

Student-Produced Response (Grid-In)

If $3a - 2b = 10$ and $a = 4$, what is the value of b ?

Question 98

Skill: Probability - Basic | Difficulty: Medium

A teacher has 5 red markers and 3 blue markers in a drawer. She randomly selects one marker, replaces it, and then selects a second marker. What is the probability that both markers are blue?

- A) $\frac{3}{8}$
 - B) $\frac{6}{64}$
 - C) $\frac{3}{32}$
 - D) $\frac{9}{64}$
-

=====

MODULE 2 - HARD ADAPTIVE (49 Questions)

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READING & WRITING - HARD MODULE

Questions 99-125 | 32 Minutes

Question 99

Skill: Inference / Complete the Text | Difficulty: Hard

Linguist Yuko Takahashi has proposed that the tonal systems of unrelated language families-such as Sino-Tibetan and Niger-Congo-may have arisen through parallel evolutionary pressures rather than common ancestry. Takahashi argues that in environments where communities relied heavily on long-distance vocal communication, tonal distinctions offered a way to increase the information density of speech without increasing volume. This hypothesis remains controversial because it is difficult to reconstruct the acoustic environments of prehistoric communities, but preliminary computational models of sound propagation in dense tropical forests _____ Takahashi's claim by showing that tonal contrasts are more easily perceived than segmental contrasts at distances exceeding 50 meters.

Which choice most logically completes the text?

- A) refute
 - B) complicate
 - C) support
 - D) disregard
-

Question 100

Skill: Inference / Complete the Text | Difficulty: Medium

The following text is adapted from Henry James's 1881 novel *The Portrait of a Lady*. The narrator describes the protagonist, Isabel Archer, shortly after her arrival in England.

She carried within her a great fund of life, and her deepest enjoyment was to feel the continuity between the movements of her own soul and the agitations of the world. For this reason she was fond of seeing great crowded streets, of pausing before shop windows, of pushing her way through knots of loungers; she liked all of it because it told her something about the world-gave her, so to speak, a taste of the world's immensity.

Based on the text, which choice best describes Isabel's outlook?

- A) She is overwhelmed by the complexity of the world and retreats into private reflection.
- B) She is drawn to bustling environments because they satisfy her appetite for experience and connection.
- C) She views crowded streets primarily as inconveniences that slow her progress.
- D) She believes that observing others is more fulfilling than directly participating in events.

Question 101

Skill: Inference / Complete the Text | Difficulty: Hard

Neuroscientist Alina Petrova and her team studied the neural mechanisms underlying "tip-of-the-tongue" states-moments when a person is certain they know a word but cannot retrieve it. Using functional MRI, Petrova observed that during such states, regions associated with phonological retrieval showed reduced activation compared to successful recall, while regions associated with semantic processing remained equally active. Petrova concluded that tip-of-the-tongue states likely arise from a specific failure in phonological access rather than a general breakdown in memory retrieval. However, she acknowledged a limitation: because participants were lying motionless in an MRI scanner, the experimental conditions differed substantially from _____ making it unclear whether the same neural patterns occur during spontaneous conversation.

Which choice most logically completes the text?

- A) natural language use,
- B) other neuroimaging studies,
- C) the participants' daily routines,
- D) earlier research on memory consolidation,

Question 102

Skill: Cross-Text | Difficulty: Hard

Text 1

Art historian Miriam Okafor contends that the paintings of the Dutch Golden Age-particularly still lifes and domestic interiors-should be understood primarily as celebrations of bourgeois prosperity. The meticulous rendering of porcelain, silverware, and imported textiles, Okafor argues, reflects the mercantile class's desire to commemorate and display its material success.

Text 2

Art critic Julian Reyes challenges purely economic readings of Dutch Golden Age painting. Reyes notes that many still lifes include *vanitas* symbols-skulls, extinguished candles, wilting flowers-that explicitly caution viewers about the transience of earthly wealth. Rather than straightforward celebrations of prosperity, these paintings embed a moral counterpoint within their lavish surfaces.

Based on the texts, how would Reyes (Text 2) most likely characterize Okafor's interpretation (Text 1)?

- A) As largely accurate but in need of minor factual corrections regarding the dates of specific paintings
- B) As incomplete because it overlooks symbolic elements within the paintings that complicate a purely celebratory reading

- C) As persuasive because it draws on a wider range of artistic evidence than competing interpretations
- D) As outdated because recent discoveries have shown that most Dutch still lifes were painted for religious institutions
-

Question 103

Skill: Inference / Complete the Text | Difficulty: Hard

The bombardier beetle defends itself by expelling a boiling-hot chemical spray from glands at the tip of its abdomen. The spray is produced by mixing two separately stored compounds—hydroquinone and hydrogen peroxide—in a reaction chamber lined with catalytic enzymes. The resulting exothermic reaction heats the mixture to nearly 100°C and generates a pressurized burst of noxious gas. Entomologist Rafael Vásquez has argued that the beetle’s defense mechanism is an example of an evolutionary “ratchet”: once the component parts of the system (separate storage chambers, a reaction chamber, catalytic enzymes) were individually selected for other metabolic functions, natural selection could incrementally co-opt them into the integrated defense system observed today. Critics of this view contend that _____

Which choice most logically completes the text?

- A) bombardier beetles are found on every continent except Antarctica, suggesting that the defense mechanism evolved very early in the lineage.
- B) the individual components of the system would not have provided a survival advantage on their own, making a gradual evolutionary pathway implausible.
- C) the chemical compounds used in the spray are commonly found in other insect species that do not possess a similar defense mechanism.
- D) the catalytic enzymes in the beetle’s reaction chamber are structurally similar to enzymes found in unrelated organisms.
-

Question 104

Skill: Words in Context | Difficulty: Medium

Economists studying the “resource curse”—the paradox that countries rich in natural resources often experience slower economic growth than resource-poor countries—have proposed several explanations. Political scientist Nadia Osman argues that resource wealth enables authoritarian governments to fund patronage networks, _____ democratic accountability. Economist Wei Chen, by contrast, emphasizes the macroeconomic mechanism: resource exports drive up a country’s exchange rate, making its manufactured goods less competitive internationally—a phenomenon known as “Dutch disease.”

Which choice completes the text with the most logical and precise word or phrase?

- A) strengthening
- B) reflecting
- C) measuring
- D) undermining
-

Question 105

Skill: Grammar / Conventions | Difficulty: Medium

In a 2024 study, archaeologist Dmitri Volkov analyzed residue found inside ceramic vessels excavated from a 4,000-year-old settlement in Central Asia. Chemical analysis revealed the presence of casein, a protein found in milk, as well as traces of fermentation byproducts. Volkov concluded that the vessels had been used to produce fermented dairy products such as cheese or yogurt. However, biochemist Lena Olsson has noted that casein can also be present in certain plant-based adhesives that ancient peoples used to seal ceramic _____ raising the possibility that the residue reflects the vessel’s construction rather than its contents.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) containers,
 - B) containers;
 - C) containers
 - D) containers-
-

Question 106

Skill: Inference / Complete the Text | Difficulty: Hard

Cognitive psychologist Raj Patel designed an experiment to test whether bilingual individuals switch between languages more efficiently when the two languages share a similar grammatical structure. Patel recruited participants fluent in both Spanish and Italian (structurally similar) and participants fluent in both Spanish and Mandarin (structurally dissimilar). He measured reaction times as participants alternated between languages in a sentence-completion task. The results showed no significant difference in switching speed between the two groups, leading Patel to conclude that _____

Which choice most logically completes the text?

- A) grammatical similarity between languages does not meaningfully affect the speed of language switching in bilingual speakers.
 - B) bilingual speakers of Spanish and Italian are more proficient overall than bilingual speakers of Spanish and Mandarin.
 - C) the sentence-completion task was too simple to produce meaningful differences in reaction time.
 - D) monolingual speakers would perform comparably to bilingual speakers on the same task.
-

Question 107

Skill: Grammar / Conventions | Difficulty: Hard

The Voynich manuscript, a fifteenth-century text written in an undeciphered script and filled with mysterious botanical illustrations, has resisted all attempts at translation. Some scholars have argued that the text is a deliberate hoax—a meaningless string of symbols designed to deceive. Computational linguist Amara Osei, however, used statistical analysis to show that the text’s character distribution patterns are consistent with those of known natural _____ a finding that a random fabrication would be extremely unlikely to replicate.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) languages,
 - B) languages:
 - C) languages
 - D) languages-
-

Question 108

Skill: Words in Context | Difficulty: Medium

In Emily Dickinson’s poetry, dashes function not merely as pauses but as structural elements that create ambiguity and force the reader to consider multiple syntactic pathways simultaneously. Literary scholar Marcus Hale has argued that this technique anticipates the fragmented syntax of modernist poets by several decades. Hale acknowledges, however, that Dickinson’s manuscripts show considerable variation in dash length and placement, and that some of the dashes in published editions may reflect editorial decisions rather than _____ intentions.

Which choice completes the text with the most logical and precise word or phrase?

- A) commercial
 - B) collaborative
 - C) authorial
 - D) pedagogical
-

Question 109

Skill: Transitions | Difficulty: Medium

Although satellite imagery has revolutionized the study of deforestation by allowing researchers to monitor forest cover across vast areas, ecologist Priya Desai cautions that satellite data alone can be misleading. In regions where secondary forest-young trees that grow after an area has been cleared-rapidly replaces primary forest, satellite images may register stable or even increasing canopy cover. _____ such images can mask a significant decline in biodiversity, because secondary forests support far fewer species than the old-growth ecosystems they replace.

Which choice completes the text with the most logical transition?

- A) In other words,
 - B) For this reason,
 - C) On the contrary,
 - D) As a result,
-

Question 110

Skill: Words in Context | Difficulty: Medium

Philosopher Hannah Arendt distinguished between “labor” (repetitive biological maintenance), “work” (the fabrication of durable objects), and “action” (engagement in the public political sphere). Arendt argued that modern industrial society had collapsed these categories, reducing all human activity to the logic of production and consumption. Political theorist Kwame Asante has extended Arendt’s framework, proposing that digital platforms have further blurred the boundary between action and labor by transforming political _____ into a continuous, algorithm-driven stream of content creation and engagement that more closely resembles biological maintenance than deliberative public discourse.

Which choice completes the text with the most logical and precise word or phrase?

- A) participation
 - B) abstention
 - C) regulation
 - D) censorship
-

Question 111

Skill: Grammar / Conventions | Difficulty: Medium

Renaissance polymath Leon Battista Alberti is credited with formulating the first systematic theory of linear perspective in painting, published in his 1435 treatise *De pictura*. Art historian Yuki Tanaka has argued that Alberti’s theory did not emerge in isolation but drew substantially on the optical research of the medieval Arab scholar Ibn al-Haytham, whose *Kitab al-Manazir* (Book of Optics) had been translated into Latin two centuries _____ making it available to European scholars well before Alberti’s time.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) earlier,
 - B) earlier;
 - C) earlier-
 - D) earlier
-

Question 112

Skill: Words in Context | Difficulty: Hard

The concept of “ecosystem services”-the benefits that humans derive from natural systems, such as clean water, pollination, and flood control-has become a central framework in environmental policy. Economist Helena Strand argues that assigning monetary values to these services, while useful for

engaging policymakers, risks _____ the intrinsic value of ecosystems by implying that nature's worth is contingent on its utility to humans.

Which choice completes the text with the most logical and precise word or phrase?

- A) enhancing
 - B) quantifying
 - C) diminishing
 - D) confirming
-

Question 113

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

- *Beloved* (1987) is a novel by Toni Morrison inspired by the true story of Margaret Garner.
- The novel uses nonlinear narration, shifting between past and present.
- Nonlinear narration is a storytelling technique in which events are presented out of chronological order.
- Morrison described this technique as essential to capturing how trauma disrupts memory.
- Literary critic Harold Bloom called *Beloved* the central work of the Western literary canon in the late twentieth century.

The student wants to define nonlinear narration and explain why Morrison employed it. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) *Beloved*, which was inspired by the true story of Margaret Garner, was called the central work of the late twentieth-century Western literary canon by Harold Bloom.
 - B) Nonlinear narration presents events out of chronological order; Morrison described this technique as essential to *Beloved* because it captures how trauma disrupts memory.
 - C) Toni Morrison's 1987 novel *Beloved* shifts between past and present throughout its narrative.
 - D) Harold Bloom praised *Beloved* as the central work of the late twentieth-century Western literary canon.
-

Question 114

Skill: Grammar / Conventions | Difficulty: Medium

Microbiologist Fatima Al-Rashidi discovered that a species of extremophile bacteria thriving in the hyper-saline waters of the Dead Sea produces a novel class of enzymes capable of functioning at salt concentrations that would denature most known proteins. Al-Rashidi has proposed that these enzymes could have biotechnological applications in industries where high-salinity conditions are _____ such as textile dyeing and petroleum extraction, where conventional enzymes rapidly lose their catalytic activity.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) unavoidable:
 - B) unavoidable-
 - C) unavoidable;
 - D) unavoidable,
-

Question 115

Skill: Inference / Complete the Text | Difficulty: Hard

The following text is adapted from Fyodor Dostoevsky's 1866 novel *Crime and Punishment*. The protagonist, Raskolnikov, wanders through the streets of St. Petersburg.

He walked along not noticing his way. The heat in the street was terrible: and the airlessness, the bustle and the plaster, scaffolding, bricks, and dust all about him, and that special Petersburg stench-all worked painfully upon the young man's already overwrought nerves.

Based on the text, which choice best describes the effect of the setting on Raskolnikov?

- A) The urban environment intensifies his already strained mental state.
 - B) The construction activity reminds him of productive labor he admires.
 - C) The heat of the city provides him with a welcome distraction from his thoughts.
 - D) The street scene calms his nerves by offering a sense of routine normalcy.
-

Question 116

Skill: Inference / Complete the Text | Difficulty: Hard

Glaciologist Maria Strand has challenged the prevailing model of ice-sheet collapse, which predicts that warming ocean water will gradually melt the undersides of coastal glaciers, causing them to thin and retreat. Strand's research on West Antarctic glaciers reveals a more complex mechanism: as meltwater pools on the glacier's surface and seeps into existing fractures, the hydraulic pressure of the trapped water can force the cracks to propagate downward through hundreds of meters of ice—a process known as hydrofracture. Strand argues that hydrofracture could trigger abrupt calving events that are not captured by gradual-melting models, meaning that current projections of sea-level rise may _____

Which choice most logically completes the text?

- A) overestimate the rate at which sea levels will rise in the coming decades.
 - B) underestimate the speed and severity of ice-sheet disintegration.
 - C) accurately predict the timeline of glacial retreat in West Antarctica.
 - D) be irrelevant to policymakers focused on coastal flood prevention.
-

Question 117

Skill: Words in Context | Difficulty: Medium

Sociologist Tomás Rivera studied patterns of civic engagement in mid-sized American cities and found that neighborhoods with higher densities of locally owned businesses tend to have higher rates of voter turnout and volunteerism. Rivera hypothesized that local business owners serve as “social anchors” who facilitate community interactions and foster a sense of shared investment in the neighborhood's _____. He acknowledged, however, that the correlation could also reflect a reverse causal pathway: communities with strong preexisting civic cultures may simply be more likely to support local businesses.

Which choice completes the text with the most logical and precise word or phrase?

- A) profitability.
 - B) uniformity.
 - C) well-being.
 - D) isolation.
-

Question 118

Skill: Grammar / Conventions | Difficulty: Medium

In astrophysics, the “faint young Sun paradox” refers to the puzzling evidence that liquid water existed on Earth's surface roughly four billion years ago, even though the Sun at that time emitted approximately 30% less energy than it does today. Under current atmospheric conditions, such reduced solar output would have left Earth entirely frozen. Climate scientist Anya Volkov has proposed that the early atmosphere contained significantly higher concentrations of methane and carbon dioxide, creating a greenhouse effect powerful enough to maintain surface temperatures above _____. Several competing hypotheses, however, attribute the warmth to other factors, such as tidal heating from the Moon's closer orbit or reduced cloud albedo.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) freezing,
- B) freezing.
- C) freezing;
- D) freezing-

Question 119

Skill: Cross-Text | Difficulty: Hard

Text 1

Philosopher Martha Nussbaum argues that literature cultivates empathy by inviting readers to inhabit perspectives radically different from their own. Through sustained imaginative engagement with fictional characters, Nussbaum contends, readers develop a capacity for moral reasoning that abstract ethical principles alone cannot produce.

Text 2

Psychologist Paul Bloom challenges the assumption that empathy is inherently moral. Bloom argues that empathy is biased and narrow: people empathize most readily with those who resemble them, which can actually reinforce prejudice. Rather than cultivating empathy, Bloom advocates for “rational compassion”—a concern for others’ well-being that is guided by reason rather than emotional identification.

Based on the texts, Bloom (Text 2) would most likely argue that Nussbaum’s claim (Text 1) about literature and empathy is:

- A) correct in asserting that empathy enhances moral reasoning but incomplete because it does not address how empathy can be biased
- B) fundamentally flawed because literature has no measurable effect on readers’ psychological states
- C) persuasive because fictional characters are specifically designed to minimize the biases Bloom identifies
- D) overly optimistic because empathy’s inherent biases may limit literature’s ability to produce the broad moral growth Nussbaum describes

Question 120

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

- The James Webb Space Telescope (JWST) was launched in December 2021.
- It is designed to observe infrared light from distant galaxies, stars, and exoplanet atmospheres.
- In 2023, JWST detected carbon dioxide in the atmosphere of exoplanet WASP-39b.
- This was the first definitive detection of carbon dioxide in an exoplanet atmosphere.
- Astrobiologist Sara Kiang described this as “a critical step toward identifying potentially habitable worlds.”

The student wants to explain the significance of the JWST’s detection of carbon dioxide on WASP-39b. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Launched in December 2021, the James Webb Space Telescope is designed to observe infrared light from distant celestial objects.
- B) The JWST’s 2023 detection of carbon dioxide in the atmosphere of exoplanet WASP-39b—the first definitive such detection—was described by astrobiologist Sara Kiang as a critical step toward identifying potentially habitable worlds.
- C) WASP-39b is an exoplanet that has carbon dioxide in its atmosphere, as detected by the JWST.
- D) The James Webb Space Telescope has made many important discoveries since its launch, including observations of distant galaxies and exoplanets.

Question 121

Skill: Words in Context | Difficulty: Medium

Biochemist Yara Haddad's research on tardigrades-microscopic animals capable of surviving extreme conditions including near-absolute-zero temperatures, intense radiation, and the vacuum of space-has revealed that tardigrades produce a unique class of intrinsically disordered proteins (IDPs) that form glass-like matrices around cellular structures during desiccation. These matrices stabilize DNA and enzymes in a suspended state until water becomes available again. Haddad's work suggests that tardigrade IDPs could potentially be synthesized and used to _____ temperature-sensitive pharmaceuticals without refrigeration, a development that would be particularly valuable in regions lacking reliable cold-chain infrastructure.

Which choice completes the text with the most logical and precise word or phrase?

- A) manufacture
 - B) prescribe
 - C) diagnose
 - D) preserve
-

Question 122

Skill: Grammar / Conventions | Difficulty: Medium

The massive sandstone formations of Monument Valley, straddling the Arizona-Utah border, have been shaped over millions of years by differential erosion. The valley's iconic buttes-isolated, flat-topped hills with steep sides-are remnants of a once-continuous plateau. Geologist Carmen Fuentes explains that the harder cap rock atop each butte resists erosion more effectively than the softer layers _____ creating the distinctive vertical profiles that have become symbols of the American Southwest.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) beneath,
 - B) beneath;
 - C) beneath
 - D) beneath-
-

Question 123

Skill: Words in Context | Difficulty: Medium

Musicologist André Baptiste has argued that the development of bebop jazz in the 1940s was not merely an aesthetic evolution but a deliberate act of cultural resistance. By increasing harmonic complexity, accelerating tempos, and foregrounding improvisation, musicians such as Charlie Parker and Dizzy Gillespie created a form that _____ casual imitation by white swing-era bands-thereby reclaiming jazz as a form of Black artistic expression.

Which choice completes the text with the most logical and precise word or phrase?

- A) facilitated
 - B) rewarded
 - C) resisted
 - D) documented
-

Question 124

Skill: Transitions | Difficulty: Medium

Anthropologist Keiko Sato studied the architectural practices of the Haida people of the Pacific Northwest coast. Sato found that the monumental cedar longhouses built by the Haida were oriented with their entrances facing the ocean, and their interior layouts reflected a strict social hierarchy, with the highest-ranking family occupying the rear of the house. _____ the physical structure of the longhouse

was not merely functional; it served as a spatial encoding of social relationships and cosmological beliefs.

Which choice completes the text with the most logical transition?

- A) Despite this,
 - B) Similarly,
 - C) Coincidentally,
 - D) In other words,
-

Question 125

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

- *Frankenstein* (1818) by Mary Shelley is often cited as the first science fiction novel.
- The novel explores the consequences of creating artificial life without ethical foresight.
- Literary scholar Brian Aldiss defined science fiction as fiction that depicts “the search for a definition of mankind and his status in the universe.”
- Aldiss argued that *Frankenstein* is the first true science fiction novel because it grounds its fantastic premise in scientific experimentation rather than supernatural forces.
- Some scholars disagree, pointing to earlier works with speculative elements, such as Johannes Kepler’s *Somnium* (1634).

The student wants to present Aldiss’s argument and note that it is contested. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Mary Shelley’s *Frankenstein* (1818) explores the consequences of creating artificial life and is considered by many to be a foundational work of literature.
 - B) Literary scholar Brian Aldiss argued that *Frankenstein* qualifies as the first true science fiction novel because it grounds its premise in scientific experimentation, though some scholars counter that earlier works, such as Kepler’s *Somnium* (1634), also contain speculative elements.
 - C) Johannes Kepler’s *Somnium* (1634) has been cited by some scholars as a work of science fiction that predates *Frankenstein* by nearly two centuries.
 - D) Brian Aldiss defined science fiction as fiction depicting humanity’s search for self-definition—a definition that applies to many novels beyond *Frankenstein*.
-

MATH - HARD MODULE

Questions 126-147 | 35 Minutes

Question 126

Skill: Systems of Equations - Number of Solutions | Difficulty: Hard

The system of equations below has no solution. What is the value of k ?

$$kx + 6y = 15$$

$$4x + 3y = 9$$

- A) 2
 - B) 4
 - C) 6
 - D) 8
-

Question 127

Skill: Quadratic Functions - Graphs & Properties | Difficulty: Medium

GRAPH TO INSERT

A quadratic function is defined by $f(x) = 2(x - 3)^2 + 5$. What is the minimum value of $f(x)$?

- A) 5
 - B) 3
 - C) 2
 - D) 11
-

Question 128

Skill: Circle Equations | Difficulty: Medium

In the xy -plane, a circle has center $(-2, 4)$ and radius 7. Which of the following is an equation of the circle?

- A) $(x + 2)^2 + (y - 4)^2 = 7$
 - B) $(x - 2)^2 + (y + 4)^2 = 7$
 - C) $(x - 2)^2 + (y + 4)^2 = 49$
 - D) $(x + 2)^2 + (y - 4)^2 = 49$
-

Question 129

Skill: Quadratic Functions - Graphs & Properties | Difficulty: Medium

If $f(x) = 3x^2 - 12x + 7$, what is the value of x at the vertex of the parabola?

- A) -4
 - B) -2
 - C) 2
 - D) 4
-

Question 130

Skill: Exponential Functions - Growth & Decay | Difficulty: Medium

A population of bacteria doubles every 3 hours. If the initial population is 500 bacteria, which function $P(t)$ models the population after t hours?

- A) $P(t) = 500(2)^{(t/3)}$
 - B) $P(t) = 500(2)^{(3t)}$
 - C) $P(t) = 500(3)^{(t/2)}$
 - D) $P(t) = 500 + 2t/3$
-

Question 131

Skill: Ratios, Rates & Proportions | Difficulty: Medium

The expression $(x^2 - 9)/(x - 3)$ is equivalent to which of the following for all $x \neq 3$?

- A) $x - 3$
 - B) $(x - 9)/3$
 - C) $x^2 - 3$
 - D) $x + 3$
-

Question 132

Skill: Right Triangle Trigonometry | Difficulty: Medium

In right triangle ABC, angle C is the right angle. If $\sin(A) = 5/13$, what is $\cos(A)$?

- A) $5/12$
 - B) $12/13$
 - C) $5/13$
 - D) $13/12$
-

Question 133

Skill: Quadratic Word Problems | Difficulty: Hard

A factory produces widgets. The profit P , in dollars, from producing x widgets per day is modeled by $P(x) = -2x^2 + 120x - 400$. What is the maximum daily profit?

- A) \$1,400
 - B) \$1,200
 - C) \$1,000
 - D) \$1,600
-

Question 134

Skill: Exponential Equations & Modeling | Difficulty: Medium

If $\log_2(x) = 5$, what is the value of x ?

- A) 10
 - B) 25
 - C) 64
 - D) 32
-

Question 135

Skill: Quadratic Word Problems | Difficulty: Medium

A ball is launched upward from ground level. Its height h , in meters, after t seconds is given by $h(t) = -5t^2 + 30t$. After how many seconds does the ball return to the ground?

- A) 3
 - B) 5
 - C) 6
 - D) 10
-

Question 136

Skill: Discriminant & Number of Solutions | Difficulty: Hard

In the equation $2x^2 + bx + 8 = 0$, for what value of b does the equation have exactly one real solution?

- A) -8
 - B) 6
 - C) 4
 - D) -4
-

Question 137

Skill: Ratios, Rates & Proportions | Difficulty: Medium

The function g is defined by $g(x) = (2x + 1)/(x - 4)$. For what value of x is $g(x)$ undefined?

- A) -4
- B) $-1/2$
- C) $1/2$
- D) 4

Question 138

Skill: Function Notation & Evaluation | Difficulty: Medium

A company's revenue R (in thousands of dollars) and advertising spending A (in thousands of dollars) are related by $R = 15\sqrt{A}$. If the company spends \$16,000 on advertising, what is the revenue, in thousands of dollars?

- A) 60
 - B) 45
 - C) 30
 - D) 240
-

Question 139

Skill: Triangle Properties | Difficulty: Medium

In a right triangle, one leg has length 8 and the hypotenuse has length 17. What is the length of the other leg?

- A) 9
 - B) 12
 - C) 15
 - D) 25
-

Question 140

Skill: Conditional Probability & Two-Way Tables | Difficulty: Hard

The table below shows the results of a survey of 150 employees about their commute method.

	Drive	Public transit	Total
Under 30 min	40	35	75
30 min or more	45	30	75
Total	85	65	150

If an employee is randomly selected from those who use public transit, what is the probability that the employee's commute is under 30 minutes?

- A) $35/65$
 - B) $35/75$
 - C) $35/150$
 - D) $30/65$
-

Question 141

Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Medium

If the function $h(x) = ax^3 - 4x + 7$ and $h(1) = 6$, what is the value of a ?

- A) -3
 - B) 1
 - C) 3
 - D) 5
-

Question 142

Skill: Quadratic Equations - Solving | Difficulty: Medium

Student-Produced Response (Grid-In)

What is the positive solution to the equation $x^2 - 5x - 14 = 0$?

Question 143

Skill: Function Transformations | Difficulty: Medium

The graph of $y = f(x)$ is a parabola with vertex at $(3, -2)$. The graph is then translated 4 units to the left and 5 units up. What is the vertex of the translated parabola?

- A) $(7, 3)$
 - B) $(7, -7)$
 - C) $(-1, -7)$
 - D) $(-1, 3)$
-

Question 144

Skill: Systems of Linear Equations | Difficulty: Hard

For the system of equations:

$$3x - 2y = 7$$

$$6x - 4y = k$$

For what value of k does the system have infinitely many solutions?

- A) 7
 - B) 14
 - C) -14
 - D) 3.5
-

Question 145

Skill: Volume & Surface Area | Difficulty: Hard

A cone has a volume of 48π cubic inches and a height of 9 inches. What is the radius of the base, in inches? (Volume of a cone = $(1/3)\pi r^2 h$)

- A) 4
 - B) 8
 - C) 16
 - D) 48
-

Question 146

Skill: Polynomial Operations | Difficulty: Medium

Student-Produced Response (Grid-In)

If $(2x + 3)(x - 5) = 2x^2 - 7x + c$, what is the value of c ?

Question 147

Skill: Exponential Functions - Growth & Decay | Difficulty: Medium

Student-Produced Response (Grid-In)

The function f is defined by $f(x) = 3(2)^x$. What is the value of $f(4) - f(2)$?

END OF TEST
